

Discrete Kantorovich Formulation

Exercise statements

This version contains the exercises only. Hints and worked solutions are available in the companion PDF.

Exercise 1 (1D discrete $\Rightarrow$ optimal Kantorovich plan).

Let $n = m = 3$, $x_1 < x_2 < x_3$, $y_1 < y_2 < y_3$, and consider two finite measures of mass 3 with weights $a = (1.1, 0.8, 1.1)$ and $b = (1, 1, 1)$. For the quadratic cost $c(x, y) = |x - y|^2$, compute the optimal coupling matrix $P \in \mathbb{R}_+^{3 \times 3}$.

Exercise 2 (Parametric 1D cases).

In both parts use strictly increasing source and target supports and the quadratic cost $|x - y|^2$. The weights below have equal total mass but are not normalized; dividing both marginals and the resulting plan by that mass gives probability measures.

- (1) Two against two: $a = (1 + \varepsilon, 1 - \varepsilon)$ and $b = (1, 1)$ with $|\varepsilon| \leq 1$. Compute the optimal $P \in \mathbb{R}_+^{2 \times 2}$ as a function of the sign of ε .
 - (2) Three against three: $a = (1 + \varepsilon, 1 + \eta, 1 - \varepsilon - \eta)$ and $b = (1, 1, 1)$ with feasibility $1 + \varepsilon \geq 0$, $1 + \eta \geq 0$, $1 - \varepsilon - \eta \geq 0$. Compute the optimal monotone $P \in \mathbb{R}_+^{3 \times 3}$ in the two regimes (i) $\varepsilon > 0$, $\eta > 0$ and (ii) $\varepsilon < 0$, $\eta > 0$.
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Exercise 3 (Couplings in basic cases). (a) What are the couplings between δ_x and δ_y ?

- (b) What are the couplings between δ_x and $\frac{1}{2}(\delta_y + \delta_z)$, with $y \neq z$?
 - (c) Let α, β be finite *positive* measures with *different* total masses. What is $\Pi(\alpha, \beta)$, the set of couplings?
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Exercise 4 (Discrete α_t from a coupling).

Let $\alpha_0 = \sum_{i=1}^n a_i \delta_{x_i}$ and $\alpha_1 = \sum_{j=1}^m b_j \delta_{y_j}$ be probability measures on $\mathbb{R}^d$ ($a_i, b_j > 0$, $\sum_i a_i = \sum_j b_j = 1$). Assume the x_i are distinct and the y_j are distinct. Let π be a quadratic-cost optimal coupling between α_0 and α_1 and write it in discrete form

$$\pi = \sum_{i=1}^n \sum_{j=1}^m P_{i,j} \delta_{(x_i, y_j)}, \quad P \in \mathbb{R}_+^{n \times m}, \quad \sum_j P_{i,j} = a_i, \quad \sum_i P_{i,j} = b_j.$$

For $t \in [0, 1]$ define $I_t : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ by $I_t(x, y) = (1 - t)x + ty$ and set

$$\alpha_t := (I_t)_\# \pi.$$

- (a) Show that α_t interpolates between α_0 and α_1 , i.e. $\alpha_0 = (P_0)_\# \pi$ and $\alpha_1 = (P_1)_\# \pi$.
- (b) Prove that α_t is discrete for every $t \in [0, 1]$. Give an upper bound on the number of Dirac masses of α_t in terms of P . Deduce the crude bound $\#\text{supp}(\alpha_t) \leq nm$. (Bonus: explain why one can choose an *optimal* coupling with at most $n + m - 1$ nonzeros, hence an interpolation with at most $n + m - 1$ atoms.)
- (c) Give the explicit expression of α_t in terms of the matrix $P \in \mathbb{R}_+^{n \times m}$.
- (d) Assume $n = m$. Under which condition(s) can one choose an optimal coupling so that α_t has exactly n Dirac masses for all t ? Give a concrete example where this happens and write α_t explicitly.

Further exercises

Exercise 5 (When does a finite-cost coupling exist?).

Let $n \in \mathbb{N}$ and consider the uniform discrete measures

$$\alpha = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}, \quad \beta = \frac{1}{n} \sum_{j=1}^n \delta_{y_j}.$$

Let $C = (C_{i,j}) \in (\mathbb{R} \cup \{+\infty\})^{n \times n}$ be a cost matrix; $+\infty$ means the pair (x_i, y_j) is forbidden. A coupling π has *finite cost* if it assigns zero mass to all $+\infty$ entries.

(i) **2×2 by hand.** For

$$C = \begin{pmatrix} C_{1,1} & C_{1,2} \\ C_{2,1} & C_{2,2} \end{pmatrix},$$

determine exactly when a finite-cost coupling exists in terms of the positions of $+\infty$. When it exists, write one explicit coupling matrix $P \in \mathbb{R}_+^{2 \times 2}$ with row/column sums $1/2$ supported only on finite entries.

(ii) **General uniform case.** Let $E = \{(i, j) : C_{i,j} < +\infty\}$ be the set of allowed edges. Prove:

There exists a finite-cost coupling

$$\iff \exists \sigma \text{ a permutation with } (i, \sigma(i)) \in E \text{ for every } i.$$